

Hi! My name is Lilly George.



I am a student in **Purdue University's John Martinson Honors College** studying **Mechanical Engineering** with a minor in **Materials Science and Engineering**.

Experiences working with **Honda** and **Purdue Electric Racing**, as well as performing **Undergraduate Research** have sparked my interest in **Motorsports, Product Design**, and **Product Development**.

Honda

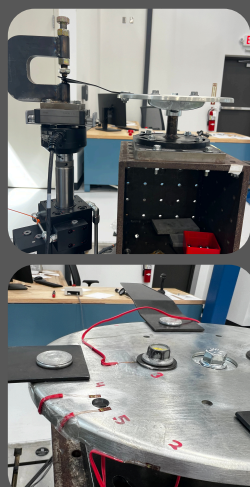
Hydraulically Actuated Damage Test

Gathered strain and acceleration data from a lawnmower being put through a durability test. Sensors placed in areas only sensitive to vertical loading. Wrote MATLAB script to calculate damage accrued using rainflow cycle counting. Correlated logged data to vertical movement of hydraulic actuators under lawnmower wheels, calibrated using damage. Hydraulic test setup will be used to decrease test time and take parts to failure.



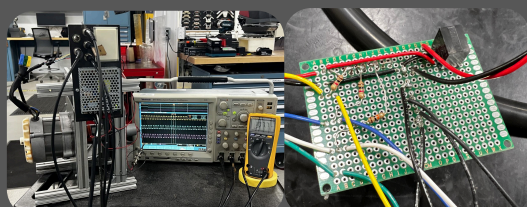
Serviceability Improvement

Initiated countermeasure for serviceability issue by requesting movement and size change of blade plate drain holes to allow for effective torquing of blade plate bolts. Updated CAD in Catia V5, presented change to design team, and validated change with FEA and strain gauges in stress test.



Electronics Test Bench

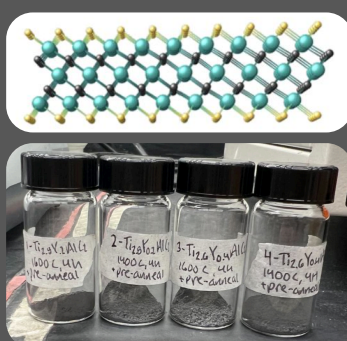
Built stand to drive electric motors without associated controller. Created circuit board to power drive motors and read hall sensor and thermistor outputs. Test bench can test all electronic motors used on Honda lawnmowers.



Anasori Lab

Rare Earth Metal Doped MXenes

Investigating feasibility and characteristics of Yttrium-doped $\text{Ti}_3\text{C}_2\text{MX}_2$ MXenes for semiconductive applications. Optimizing firing, etching, and intercalation steps while measuring conductivity and magnetic properties. Project ongoing: My PI has kindly let me continue this project this summer through the help of in-person lab members. I am managing progress from afar and planning next steps as tasks are completed.



Training

Glovebox, fumehood, centrifuge, HF/HCl acid, lab safety



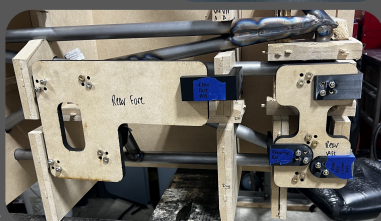
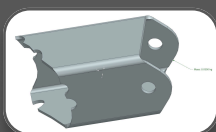
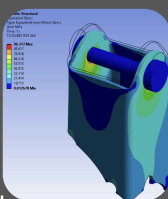
Click Here!

Purdue Electric Racing

FSAE Electric - Vehicle Dynamics Team - Suspension/Steering/Brakes

Control Arm Tab Design

CAD, cut normalization, FEA, manufacturing, and assembly of 16 tabs that connect 2026 car's control arms to the chassis. Required detailed CAD while coping tabs to frame, FEA, using a rotary tube laser, and MDF boards/3D prints to create installation jigs.



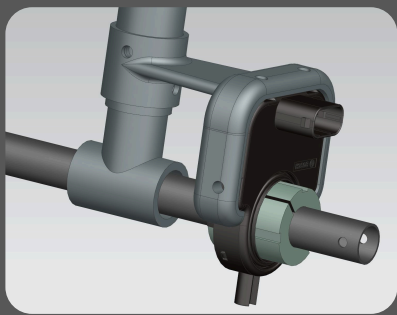
UTM Suspension Testing

Worked with professor to safely design, plan, and execute testing of mock suspension members in uniaxial tension. Successfully broke and compared seven different specimens. Data used for future design of carbon fiber twill tube suspension members.



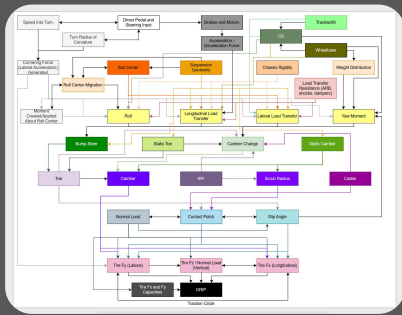
Sensor Mount

First part designed for 2025 car. Design utilized steering column supports and set screw holes to support steering angle sensor while on car. 3D printed.



Vehicle Dynamics Map

Supplemented a vehicle kinematics presentation. Aided in team member visualization of links between aspects of vehicle dynamics.

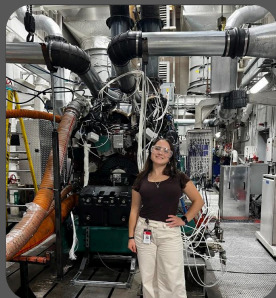


Cummins

Summer '25 Mechanical Engineering Intern Power Systems Air Handling/Integration Team

Engine Program:

Worked with team members in the US, UK, and India to determine necessary validation tasks to push engine program to production phase.



Turbomachinery Training:

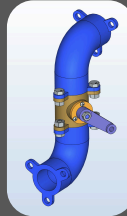
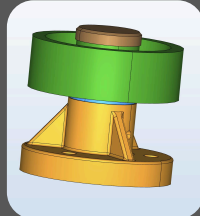
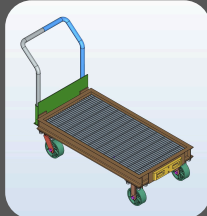
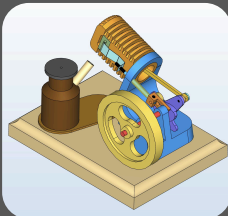
Lessons on inner workings, design, specifications, trade offs, and failure modes of turbochargers.



College Coursework

MFET 16300

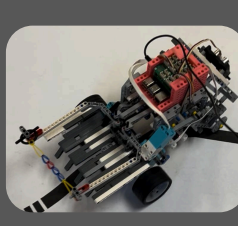
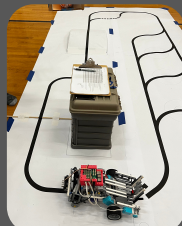
Class with use of Siemens NX, Teamcenter, and Anark Collaborate. Included drafting and assembling a "Little Blazer Engine", a spindle, and a butterfly valve.



HONR ENGR 161/162 (First Year Engineering)

Mars Rover Robot Design

Task: Build rover to hold/carry/dispense cargo while navigating terrain using given sensors, motors, Legos, and Raspberry Pis. My focuses: sensor use/placement, strength for terrain navigation



Disaster Relief Robot Design

Task: Assemble a robot to carry relief supplies through disaster zones, using materials given in Mars Rover project above. Successes: maze navigation, path mapping, dispensing cargo, hazard identification/avoidance/mapping. My focuses: structural integrity, cargo dispensing, sensor use

